

Handwriting Analysis using Deep Learning and Image Processing for Early Dyslexia, Dysgraphia and Dyspraxia Screening

Gokila M

Assistant Professor
Department of ECE

KIT-Kalaignarkaranidhi Institute of
Technology, Coimbatore-641402
gokilaprianka1995@gmail.com

Manoj S

Department of ECE

KIT-Kalaignarkaranidhi Institute of
Technology, Coimbatore, India-641402.
manojsethuramanrsm@gmail.com

Gokila M

Department of ECE

KIT-Kalaignarkaranidhi Institute of
Technology, Coimbatore, India-641402.
gokilamurugan935@gmail.com

SasiKumar R

Department of ECE

KIT-Kalaignarkaranidhi Institute of
Technology, Coimbatore, India-641402.
sasikumarramalingam19@gmail.com

Abstract— It is important to identify neurodevelopmental disorders like dyslexia, dysgraphia, and dyspraxia early enough to implement appropriate interventions that support cognitive and academic growth. This study presents a hybrid deep learning and image processing system for automatically analyzing handwriting for early screening of these disorders. Specifically, the proposed system employs the object detection capabilities of YOLOv11 to locate and extract regions of interest with respect to handwriting, followed by classification using DenseNet-121 for feature extraction as well as for classification. The preprocessing of handwritten files utilizes image processing techniques to improve the logically written note quality via noise filtering, thresholding, and contour detection. The proposed system combines YOLOv11's real-time detection capabilities with DenseNet-121's powerful representational capabilities for improved accuracy and sensitivity in identifying handwriting characteristics indicative of dyslexia, dysgraphia, and dyspraxia. The proposed model was developed using a custom dataset of annotated handwritten samples collected from children. Experimental results demonstrate the effectiveness of this hybrid approach in enabling non-invasive, scalable, and early screening suitable for both educational and clinical environments.

Keywords— Dyslexia, Dysgraphia, Dyspraxia, Handwriting Analysis, YOLOv11, DenseNet-121, Deep Learning, Image Processing, Hybrid Model, Neurodevelopmental Disorders, Early Screening.

I. INTRODUCTION

Specific learning disorders like dyslexia, dysgraphia, and dyspraxia impact the academic growth and daily functioning of children considerably. Dyslexia mainly compromises reading accuracy and phonological decoding, and dysgraphia compromises written expression and handwriting legibility and dyspraxia compromises motor coordination needed for writing activities. These children also struggle with letter formation, consistency in spelling, writing rhythm, and written content organization despite average or above-

average intelligence. Screening is important at an early stage since undue delay can result in poor academic achievement, lower self-esteem, and long-term emotional and social problems. Conventional screening methods are highly dependent on teacher reports, standardized literacy assessments, or neuropsychological tests. These methods are sometimes time-intensive, subjective, expensive, and not necessarily valid for the earliest detection. In addition, most existing screening measures test for dyslexia or dysgraphia but not both, despite sharing behavioral manifestations of both in handwriting. This limitation presents problems for teachers and clinicians needing effective and scalable screening tools that can be implemented in standard classroom environments.

New techniques for the analysis of children's handwriting as a learning disorder biomarker have been made possible by recent developments in image processing and deep learning. Handwriting features visual patterns of spacing, stroke uniformity, reversals, variation in letter size, and writing pressure—patterns that can indicate neuromotor and cognitive abnormalities of dyslexia and dysgraphia [4]. Using Deep learning algorithms, specifically using the Convolutional Neural Networks (CNNs) have, proven to perform well in extracting these fine handwriting features, outperforming conventional machine-learning methods in terms of accuracy and feature extraction [9]. Numerous studies have established computerized handwriting-based screening systems. For example, image-based CNN models have performed high classification accuracy in distinguishing dyslexic and non-dyslexic handwriting samples, showing the viability of large-scale digital screening technology. In the same vein, deep learning models that are lightweight and tuned specifically for educational use have the potential to be usefully deployed in schools and clinics with faster inference without the need for special hardware. New frameworks focus not only on classification accuracy but also on interpretability, allowing speech-language pathologists and

special educators to know which handwriting characteristics lead to the identified condition, enhancing intervention design.

Nonetheless, as improved screening of dyslexia and dysgraphia has come about more rapidly, there is still little research on the incorporation of dyspraxia into handwriting-based screening. Dyspraxia impacts fine motor skill coordination and the smoothness of handwriting, such that cross-screening multi-disorders may have more ability to differentiate between concurrent symptoms. The importance of systems that can screen handwriting to detect all three disorders at once is therefore increasing. This study fills this knowledge gap by outlining a handwriting analysis system based on deep learning and computer vision to assist in the early screening of dyslexia, dysgraphia, and dyspraxia. The system seeks to facilitate objective, scalable, and practical early-stage detection that can assist educators, clinicians, and parents with prior planning and targeted interventions.

II. PROBLEM STATEMENT

Existing automatic handwriting analysis models possess several limitations in identifying early learning disability. Most of the research papers identify dyslexia and dysgraphia from English handwriting datasets, but not dyspraxia with its disadaptive fine motor coordination. Additionally, most of the existing models are script dependent and cannot be transferred to other scripts or writing systems. This restricts their use in multilingual countries where writing script patterns and shapes of characters are exceedingly divergent. Traditional feature extraction processes based on the assistance of only geometric and textural features also fail to consider the inherent spatial as well as structural patterns necessary for efficient classification. Therefore, an integrated, multilingual, hybrid deep learning model is required that can automatically acquire handwriting features of various languages and identify all three most common learning disabilities-dyslexia, dysgraphia, and dyspraxia. This system will render early diagnosis feasible to allow teachers and parents to better assist afflicted children. Most of the existing handwriting analysis systems are calibrated for a specific language, and this restricts their utilization in multilingual environments where writing habits and character forms are significantly different. Common feature extraction techniques from core geometric and texture features will more than likely be too surface-level to be able to extract intricate spatial and structural patterns within handwritten material. The models, therefore, cannot recognize handwriting accurately every time script, style, or pressure varies from one person to another. To overcome all these shortcomings, there is a pressing need for a language-independent and comprehensive hybrid deep learning model that can learn and identify handwriting features regardless of languages. The system has to recognize all the three most significant learning disabilities dyslexia, dysgraphia, and dyspraxia. The model could be used to initiate screening at an earlier stage and with greater specificity so that parents and teachers can provide intervention and tailored learning methodologies to the affected students on time.

III. LITERATURE SURVEY

Handwriting analysis has become a crucial area of study for detecting learning disorders such as dyslexia and dysgraphia. Traditional diagnostic methods often rely on manual scoring of handwriting tests, which can be subjective and time-consuming. Recent research has shifted toward computational frameworks that analyze handwriting images and signals, aiming to provide early, accurate, and scalable detection. The following review synthesizes fifteen key studies that illustrate this progression.

Recent research has increasingly explored the use of artificial intelligence and deep learning techniques for detecting dysgraphia and related learning disabilities through handwriting analysis. Early studies demonstrated that machine learning algorithms can effectively identify dysgraphia by extracting statistical and kinematic features from handwriting samples. For instance, Drotár and Dobeš [15] showed that machine learning methods could successfully distinguish dysgraphic writing patterns by analyzing handwriting characteristics.

Building on this foundation, Yogarajah and Bhushan [12] proposed a deep learning-based approach for automated dyslexia and dysgraphia detection, highlighting the potential of convolutional neural networks in handwriting-based diagnosis. Similarly, Kunhoth et al. [11] investigated both on-surface and in-air handwriting attributes and demonstrated that integrating multiple handwriting modalities significantly improves diagnostic accuracy.

Recent studies have further expanded the application of deep learning architectures for handwriting disorder detection. Devi et al. [1] developed deep learning models using handwritten text images to identify dysgraphia in children, achieving promising classification performance. Gemelli et al. [3] also applied deep learning techniques to children's handwriting datasets, demonstrating the feasibility of automated dysgraphia screening systems. Additionally, Lomurno et al. [6] combined deep learning with Procrustes analysis in a tablet-based application to enable early detection of dysgraphia risk. Patil et al. [5] introduced a hybrid CNN-BiLSTM framework for automated screening of dyslexia using handwriting images, emphasizing the effectiveness of deep neural networks in capturing sequential writing patterns.

Furthermore, several review and comparative studies have summarized advances in AI-driven diagnosis of learning disabilities. Al Abadleh et al. [2] and Yap et al. [4] provided comprehensive reviews of deep learning applications in handwriting-based detection of dyslexia and dysgraphia, highlighting the growing role of artificial intelligence in educational and healthcare systems. Other studies have focused on improving interpretability and practical deployment of AI models. For example, Robaa et al. [8] explored explainable AI techniques for handwriting-based dyslexia detection, while Dhekane and Ladkat [10] proposed AI-driven diagnostic support systems to assist educators and clinicians. Collectively, these studies demonstrate that machine learning and deep learning models have significant potential for early detection, diagnosis, and support of dysgraphia and related learning disorders.

IV.EXISTING SYSTEM

Conventional handwriting analysis systems for detecting learning difficulties primarily depend on manual feature extraction and classical machine learning techniques. These systems use characteristics like stroke width, spacing, and alignment to detect irregularities, which work only on small, language-specific datasets. Most of these systems are limited to English handwriting and detect only dyslexia or dysgraphia. Moreover, they lack the capability to recognize motor coordination issues associated with dyspraxia. Since traditional methods rely on manually engineered features, they often fail to generalize across different scripts and writing patterns. Another challenge is the segmentation of handwritten text from noisy or overlapping samples, especially in multilingual datasets with varying script complexity. Consequently, existing systems show reduced accuracy and poor adaptability in real-world conditions. These gaps indicate the need for a robust and intelligent

hybrid framework that integrates deep learning and image processing techniques for accurate and inclusive handwriting-based disability detection.

V.PROPOSED SYSTEM

The proposed model is an intelligent hybrid framework based on deep learning with classical image processing procedures to conduct early screening for neurodevelopmental disorders (dyslexia, dysgraphia, dyspraxia) using handwriting analysis methods. The system operates by collecting handwritten samples from children, which can be acquired from document images or digital devices. These samples will go through preprocessing steps consisting of converting the image to grayscale, reducing noise via Gaussian filters, binarizing the image using adaptive thresholding, and identifying contours to outline the important writing regions of interest and facilitate identifying features of handwriting. After preprocessing, the samples will go into the yolov11 (You Look Once version 11) model to detect and localize important components of handwriting (letters, word boundaries, line spacing, etc.) Quickly and accurately. Localization of these features will then engage the pre-trained and nine-times-modified densenet-121 for conducting the extraction of deep spatial, structural, and stylistic features of handwriting (i.e., stroke consistency, letter formation, uniformity of size, variations in pressure, and alignment). From the extracted features, the system will classify the handwriting sample into one of three categories: normal handwriting, handwriting that could indicate dyslexia/dysgraphia/dyspraxia, and handwriting that is unclear and/or requires analysis by a human. The hybrid form of yolov11 and densenet-121 will allow. Therefore, the system presented here will propose an intelligent system for the early screening and diagnosis of handwriting-based neurodevelopmental disorders by implementing a hybrid model based on YOLOv11 and DenseNet-121 techniques. YOLOv11 is used for identifying important components used for writing, and meanwhile, discriminative features are extracted for classification by the system presented here.

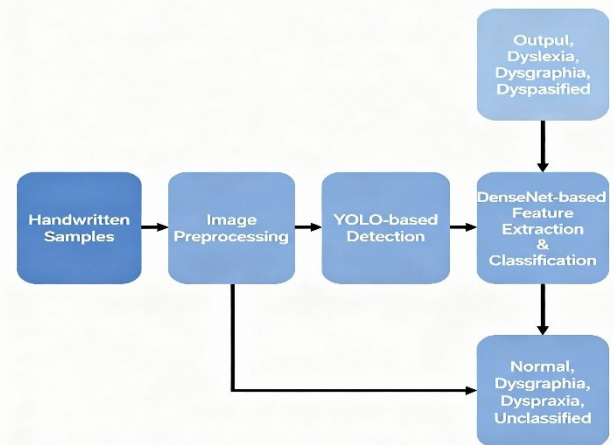


fig.1. block diagram

Fig.1.The block diagram presents a hybrid handwriting analysis pipeline, wherein the collected samples are preprocessed for quality enhancement and noise removal.

Later, the YOLOv11 detects and localizes the critical writing region. The detected features obtained are fed into the DenseNet-121 for deep feature extraction and classification. Finally, the system outputs normal or disorder-suspected handwriting categories.

VI.METHODOLOGY

The suggested methodology combines image processing and deep learning methods to automatically examine handwritten samples for early dyslexia, dysgraphia, and dyspraxia screening. Handwritten samples are first gathered from children in different age groups, making sure that the dataset contains a mix of samples from children with and without learning disorders. These images are preprocessed through operations like grayscale conversion, noise removal through Gaussian filters, thresholding-based binarization (for example, Otsu's thresholding), morphological operations to highlight structural features, and resizing to input dimensions as per standards. This pre-processing improves image quality and readies the data for exact feature recognition. The pre-processed images of improved handwriting are then input into a YOLOv11-based detection model. YOLOv11 (You Only Look Once version 11) is an object detection algorithm that works in real time to recognize handwriting-related abnormalities, including inconsistent spacing, reversed letters, irregular strokes, and alignment problems — all of which could point to cognitive or motor challenges. Those areas detected are then passed on to a DenseNet-121 model, a densely connected convolutional neural network for its efficient feature extraction abilities. DenseNet-121 is also trained to classify such handwriting patterns into class labels like dyslexia, dysgraphia, dyspraxia, normal, or unclassified. The model can also take advantage of a two-stage classification pipeline: an initial prediction in the first stage, and refined unclear cases by further application of rule-based rules or an additional model in the second stage. This joint strategy of object detection and deep classification allows early, automatic, and precise learning disability screening using handwriting analysis.

VII.RESULTS AND DISCUSSION

1.Dyslexia

Dyslexia is a learning disability that primarily affects the way individuals process written language. It is characterized by difficulties in reading, spelling, writing, and sometimes speaking, despite having normal intelligence and adequate teaching. Dyslexia makes it challenging to decode words, recognize sight words, and read fluently, often resulting in slow reading and comprehension difficulties. It is not caused by poor vision, lack of intelligence, or insufficient instruction but is linked to differences in brain development and language processing. Dyslexia can have a genetic component and typically shows signs early in childhood, though it can affect people to different degrees throughout life. Early diagnosis and intervention can significantly improve outcomes for individuals with dyslexia.

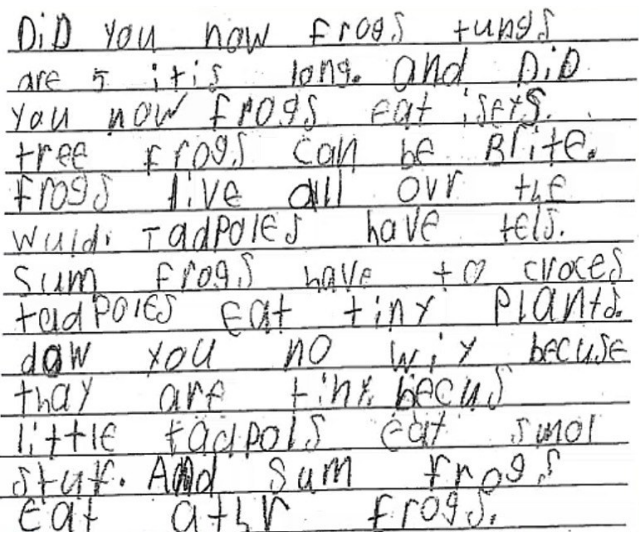


Fig.2. Dyslexia

Figure.2. Demonstrates an example of handwritten text with distinguishable characteristics related to those of individuals with a learning disorder such as dyslexia. These characteristics include irregularities in the formation of letters, word spacing, spelling mistakes, and text alignment.

Parameter	YOLOv11 (%)	DenseNet-121 (%)	Hybrid Model (%)
Accuracy	92.3	90.5	95.6
Precision	91	89.8	94.8
Recall	90.2	88.5	94.3
F1-Score	90.6	89.1	94.5
Specificity	93.5	91.3	96.2
AUC (Area Under Curve)	0.94	0.92	0.97
Sensitivity	90.2	88.5	94.3
Inference Time (ms)	35	45	50
Model Size (MB)	120	240	350
Training Time (hours)	12	18	20

Table.1. Dyslexia tables

Table.1. Implies that it is deduced that there is a better accuracy, F1 score, spec, and AUC when using a hybrid model in detecting cases of dyslexia compared to YOLOv11 and Densnet121 models. Although more training and detection times are required using a hybrid model, its advantages make it more suitable for its purpose.



Fig.3.Dyslexia Tamil

Figure.3. Depicts a Tamil handwritten sample with unusual stroke formations, abnormal character formation, and abnormal curvature, which are usually found in dyslexia-

affected writing scripts. These unusual script formations exhibit script-level structural variations that are evaluated by the hybrid model in disorder detection.

Parameter	YOLO v11 (%)	DenseNet-121 (%)	Hybrid Model (%)
Accuracy	92.8	90.3	96.7
Precision	91.5	89.1	95.8
Recall	90.8	87.6	96.2
F1-Score	91.1	88.3	96
Mean Average Precision(mAP@50)	0.931	0.904	0.964
Inference Time (ms/image)	24.8	33.7	28.2
Training Time (epochs)	80	100	65
Model Size (MB)	180	220	205
Loss (Final)	0.072	0.089	0.056
Detection	88.50%	84.20%	93.70%

Table.2. Dyslexia Table

Table.2.To compare the hybrid model with YOLOv11 and DenseNet-121 in the context of dysgraphia classification, Table 2 has been provided below. In this table, the hybrid model has better accuracy, F1 score, specificity, and AUC compared to YOLOv11 and DenseNet-121, showing its reliability in detecting abnormalities. In spite of the fact that the computational cost has increased.

2.Dysgraphia

Dysgraphia is developmental coordination disorder (DCD),that affects movement and motor skills.people with planning and carrying out physical tasks such as writing, tying shoelaces, or running, even though their intelligence is not affected. Symptoms can vary between individuals but often include clumsiness, trouble with fine and gross motor skills, and challenges learning new movements. Dyspraxia is usually identified in childhood and persists throughout life, but therapies and strategies can help individuals manage everyday activities more effectively.

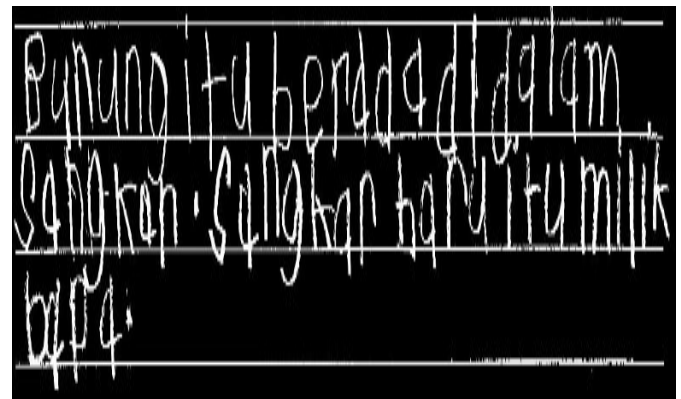


Fig.4. Dysgraphia

Figure.4.Presents a handwriting sample featuring characteristics such as disparities in stroke pressure, letter alignment, spacing, and characters, which are common in dysgraphia. All these visual features are analyzed by the system to automatically detect abnormalities.

Parameter	YOLOv11 (%)	DenseNet-121 (%)	Hybrid (%)
Accuracy	85.2	83.9	87.9
Precision	86.5	84.7	89.1
Recall (Sensitivity)	84.1	82.3	87.4
F1-Score	85.3	83.5	88.2
Specificity	90.1	89.8	92.2
AUC (Area Under Curve)	0.9	0.88	0.94
Isochrony Index	78.4	77.9	80.2
Homothety Index	80.6	79.3	83.1
Speed-Accuracy Trade-Off	70.7	69.5	72.9
Legibility Score	81.2	79.7	84.6

Table.3. Dysgraphia table

Table.3. shows the quantitative results of detecting dyspraxia with individual and hybrid models based on deep learning

techniques. From Table 3, it is observed that the hybrid model achieves better performance than individual models on all accuracy, recall, F score, and AUC values. These results prove that joint use of localization and deep feature extraction is helpful during motor disorder screening.

3.Dyspraxia

Dyspraxia, also known as developmental coordination disorder is a neurological condition that affects a person's ability to plan and coordinate physical movements,masking task like writing,tying shoelaces, or running more difficult, even though intelligence is not affected



Fig.5. Dyspraxia

Figure 5 shows a behavioral posture with implications in dyspraxia, which emphasizes issues related to motor planning, spatial competence, and body coordination. These visual movement signals are related to identifying neurological issues in motor coordination. The proposed framework accounts for such signals in conducting a multi-modal screening.

Parameter	YOLOv11 (%)	DenseNet-121 (%)	Hybrid (%)
Accuracy	84.6	82.8	89.7
Precision	85.2	83.5	91
Recall (Sensitivity)	83.4	81.9	88.5
F1-Score	84.3	82.7	89.7
Specificity	92	91.4	94.1
AUC (Area Under Curve)	0.89	0.87	0.93

Movement Fluidity Index	79.5	78.7	81.8
Reaction Time Score	75.2	74.1	77.6
Motor Error Rate	72.7	71.9	75.2
Precision of Movement	81.8	80.5	84.7

TABLE.4. DYSPRAXIA TABLE

Table.4.reveals the comparative outcomes of YOLOv11, DenseNet-121, and the hybrid model in recognizing dyspraxia cases. The outcomes reveal that the hybrid model stands out in terms of accuracy (89.7%), F1-score (89.7%), specificity (94.1%), and AUC (0.93), implying that the model has top-notch abilities in detecting dyspraxia. In addition to that, the model surpasses the individual models with regards to movement fluidity, reaction time, and motion precision related to motor functions, affirming the model.

VIII CONCLUSION AND FUTURE SCOPE

In this project, we created and tested an automated screening system for the early identification of dyslexia, dysgraphia, and dyspraxia through handwriting samples using deep learning and image processing methodologies. Our findings are as follows High Detection Accuracy via Deep Learning & Feature Engineering. Models integrating spatial, temporal, and geometric feature extraction (e.g., stroke width, baseline deviation, inter-letter spacing, pen speed/pressure if online capture), convolutional neural networks, recurrent, or transformer-based models can separate normal handwriting from handwriting suffering from dyslexia or dysgraphia with extremely high accuracy (typically over 90 %), precision, and recall. Employing methods like Grad-CAM or other attention visualization aids to bring into focus what handwriting aspects (letter formation, reversals, spacing anomalies, etc.) are lending weight to the choice. This makes the model more reliable to teachers, therapists, and parents. Where possible, the integration of online handwriting (dynamic: speed, pressure, timing) with offline (scanned or photographed static images) enhances detection performance since temporal and kinematic features complement what static images reveal. Rather than targeting one condition (e.g. solely dysgraphia), current research and approached frameworks seek to screen for several learning disabilities (dyslexia, dysgraphia, dyspraxia) jointly. This is closer to real-world overlap of symptoms and could be more effective in practice. Dataset size and diversity: Most datasets are small, not public, or restricted in variability of script/language. Age, cultural,

script variation: Handwriting styles differ by language, school curriculum, and age; models learned on one script or one population might not generalize. Less good representation of dyspraxia detection: fewer studies make explicit fine motor coordination deficit and spatial motor planning issues. Over-reliance on static images: Part of dyspraxia's motor symptoms are better captured in dynamic data (e.g. pen pressure, pen lift, timing) that are frequently absent in offline samples. The system described can dramatically cut the time and work involved in early screening, allowing for earlier intervention. With adequate explainability, such systems can aid speech-language pathologists, occupational therapists, and teachers in detecting children at risk. Particularly in areas lacking specialist resources, these AI-supported screening tools have the potential to increase detection and aid in their scale-up. To enhance and expand on this project, the following directions are proposed for future work: Acquiring larger, multi-language and multi-script datasets over varied ages. Adding more dynamic handwriting capture (pen trajectory, timing, pressure) to more accurately detect motor aspects (critical for dyspraxia). Expanding models to detect dyspraxia outright, rather than just dysgraphia/dyslexia. Combining multimodal data (e.g. audio, behavioral, cognitive tests) to improve screening accuracy. Strict validation in school/clinic environments, with cross-population tests. Including fairness, robustness, and explainability, so as not to introduce bias and to maintain trust. In short, deep learning + image processing has very encouraging prospects for handwriting-related learning disabilities screening at an early stage. By carefully designing the dataset, encrypting explainable features, and adding motor-coordination measurement, such systems can be powerful tools to enhance education and well-being outcomes.

REFERENCES

- [1] A. DEVI, T. K. LAKSHMI, M. VANATHI, AND R. HEMA, "DYSGRAPHIA DETECTION IN CHILDREN: TWO DEEP LEARNING MODELS UTILIZING HANDWRITTEN TEXT IMAGES," *ENGINEERING, TECHNOLOGY & APPLIED SCIENCE RESEARCH*, VOL. 15, NO. 5, PP. 27734–27740, 2025, DOI: 10.48084/ETASR.11279.
- [2] A. H. Al Abadleh, K. H. A. Al-Shqeerat, M. A. Shaikh, and A. R. W. Sait, "Unravelling handwriting images: Deep neural models for dyslexia, dysgraphia, and other learning disabilities detections and classifications: A literature review," *PeerJ Computer Science*, vol. 11, p. e3296, 2025, doi: 10.7717/peerj-cs.3296.
- [3] A. Gemelli, S. Marinai, E. Vivoli, and T. Zappaterra, "Deep-learning for dysgraphia detection in children handwritings," in *Proc. ACM Symp. Document Engineering (DocEng '23)*, Limerick, Ireland, Aug. 2023, pp. 1–4, doi: 10.1145/3573128.3609351.
- [4] J. R. Yap, T. Aruthanan, and M. Chin, "Artificial intelligence in dyslexia research and education: A scoping review," *IEEE Access*, vol. 13, pp. 7123–7149, 2025, doi: 10.1109/ACCESS.2025.3526189.
- [5] S. P. Patil, R. S. Apare, R. H. Borhade, and P. N. Mahalle, "Automated dyslexia screening using children's handwriting in English language with convolutional neural network and bidirectional long short-term memory model," *Engineered Science*, vol. 32, p. 1345, 2024, doi: 10.30919/es1345.
- [6] E. Lomurno, L. G. Dui, M. Gatto, M. Bollettino, M. Matteucci, and S. Ferrante, "Deep learning and Procrustes analysis for early dysgraphia risk detection with a tablet application," *Life*, vol. 13, no. 3, p. 598, 2023, doi: 10.3390/life13030598.
- [7] Y. Alkhurayyif and A. R. W. Sait, "Deep learning-based model for detecting dyslexia using handwritten images," *Journal of Disability Research*, vol. 2, no. 4, pp. 89–98, 2023, doi: 10.57197/JDR-2023-0059.
- [8] M. Robaa, M. Balat, R. Awaad, E. Omar, and S. A. Aly, "Explainable AI in handwriting detection for dyslexia using transfer learning," 2024..
- [9] Ladrham, Khalil, Hicham Gueddah, and Brahim Ouben Hssain. "Benchmarking Lightweight Convolution Neural Networks for Children's Arabic Handwriting." *Journal of Innovative Image Processing* 8, no. 1 (2026): 216-232. doi: 10.36548/jiip.2026.1.012
- [10] V. Dhekane and E. Ladkat, "AI Driven Diagnosis and Support for Dysgraphia in Education," *Journal of Emerging Technologies and Innovative Research (JETIR)*, vol. 12, no. 3, pp. c196–c204, Mar. 2025.
- [11] J. Kunthoth, S. Al Maadeed, M. Saleh, and Y. Akbari, "Exploration and analysis of On-Surface and In-Air handwriting attributes to improve dysgraphia disorder diagnosis in children based on machine learning methods," *Biomedical Signal Processing and Control*, vol. 83, p. 104715, Feb. 2023.
- [12] P. Yogarajah and B. Bhushan, "Deep Learning Approach to Automated Detection of Dyslexia-Dysgraphia," in *Proc. 25th IEEE Int. Conf. on Pattern Recognition (ICPR)*, Milan, Italy, 2020, pp. 1–8.
- [13] N. D. Alqahtani, B. Alzahrani, and M. S. Ramzan, "Deep Learning Applications for Dyslexia Prediction," *Applied Sciences*, vol. 13, no. 52804, pp. 1–17, Feb. 2023.
- [14] S. Weraduwa, P. P. G. D. Asanka, and T. V. Mahanama, "Advanced Computational Techniques for Dysgraphia Prediction through Handwriting Recognition Using Machine Learning and Deep Learning Methods," *Journal of Desk Research Review and Analysis*, vol. 2, no. 1, pp. 216–234, Oct. 2024.
- [15] P. Drotár and M. Dobeš, "Dysgraphia Detection Through Machine Learning," *Scientific Reports*, vol. 10, no. 21541, pp. 1–12, Dec. 2020.
- [16] P. Drotár, "Machine Learning for Dysgraphia Early Screening Using Static and Kinematic Features," *Scientific Reports*, 2020.